Reasenberg and Shapiro inferred the shape of the space warp. More precisely, they inferred the shape of the two-dimensional surface formed by the paths of the *Viking* radio signals. That surface was very nearly the Sun's equatorial plane, so I describe it that way here.

The shape that the team measured, for the Sun's equatorial plane, is shown in Figure 4.4 with the magnitude of the warping exaggerated. The measured shape was precisely what Einstein's relativistic laws predict—precise to within the experimental error, which was 0.001 of the actual warping, that is, a part in a thousand. Around a neutron star, the space warp is far greater. Around a black hole, it is enormously greater.

Now, the Sun's equatorial plane divides space into two identical halves, that above the plane and that below. Nonetheless, Figure 4.4 shows the equatorial plane as warped like the surface of a bowl. It bends downward inside and near the Sun, so that diameters of circles around the Sun, when multiplied by π (3.14159 . . . ), are larger than circumferences—larger, in the case of the Sun, by roughly 100 kilometers. That's not much, but it was easily measured by the spacecraft, with a precision of a part in a thousand.

How can space "bend down"? Inside *what* does it bend? It bends inside a higher-dimensional hyperspace, called "the bulk," that is not part of our universe!

Let's make that more precise. In Figure 4.4 the Sun's equatorial plane is a two-dimensional surface that bends downward in a three-dimensional bulk. This motivates the way we physicists think about our full universe. Our universe has three space dimensions (east-west, north-south, up-down), and we think of it as a three-dimensional membrane or *brane* for short that is warped in a higher-dimensional *bulk*. How many dimensions does the bulk have? I discuss this carefully